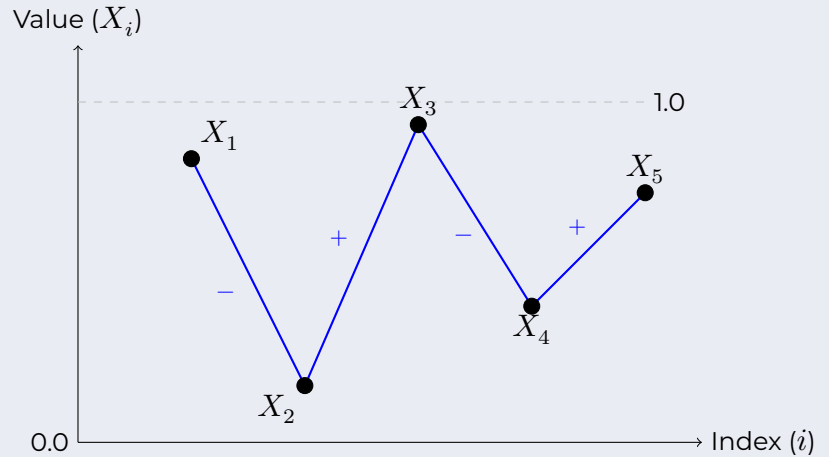


WEEKLY PROBLEM

EXERCISE.

Zigzag Subsequences

A sequence $y_1, \dots, y_k$ is *zigzag* if $k = 1$ or if consecutive differences $y_{i+1} - y_i$ are nonzero and alternate in sign. Let $X_1, \dots, X_n$ be chosen independently at random from $[0, 1]$. Let $a(X_1, \dots, X_n)$ be the length of the longest zigzag subsequence, and $f(n)$ be its expected value.



Consecutive differences: $(X_2 - X_1 < 0), (X_3 - X_2 > 0), (X_4 - X_3 < 0) \dots$
Signs: $(-, +, -, +) \Rightarrow$ **Zigzag Sequence**

Part A

1. Show any two distinct numbers form a zigzag, hence $f(2) = 2$.
2. Verify $f(3) = 8/3$ by averaging a over all $3! = 6$ equally likely orderings of three numbers.

Part B For $n \geq 3$, let $\Delta_n = a(X_1, \dots, X_n) - a(X_2, \dots, X_n)$ be the drop in length when X_1 is removed.

- (i) Show that Δ_n is always 0 or 1.
 - (ii) Show $\Delta_n = 1$ if and only if the triple (X_1, X_2, X_3) is non-monotone.
- Conclude from Part A that $\Delta_n = 1$ occurs on average $2/3$ of the time for any $n \geq 3$.

Part C Derive a formula for $f(n)$ using $f(2) = 2$ as a base case.

- (i) Explain why $f(n) - f(n-1) = E[\Delta_n]$.



(Scan this to submit solution!)